Replication of "A word-order constraint on phonological activation" by N Janssen, FX Alario, A Caramazza (2008, *Psychological Science*)

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Introduction

In their original paper "A word-order constraint on phonological activation," Janssen, Alario, and Caramazza (2008) suggest that different word-order rules create language-specific constraints that influence phonological activation by comparing French and English. Janssen et al. disagree with the prominent cascading flow model of activation, highlighting that this view is not complex enough to account for multiword production. An alternative to the unregulated-cascading hypothesis, the word-order-modulation hypothesis predicts a language-specific phonological facilitation for the second word (either noun or adjective) in the noun-adjective word pair, but no facilitation for the first word. In French, the adjective would receive facilitation because it appears second (ex. *valise vert*) in the word pair, while in English, the noun would receive facilitation (ex. *green goose*).

In this replication of Janssen et al. (2008), we sought to exactly reproduce the methodology used in the original paper. As the present work was conducted as part of the Reproducibility Project: Psychology, which asks volunteers to replicate the final experiment in their chosen paper, this is a direct replication of Experiment 2, which concerns English. Janssen et al. propose that word-order constraints modulate the activation of phonological representations and that there will be a phonological congruency effect in object naming, but not in color naming for English speakers. They found evidence to support an effect of congruency on object naming, $t(14) = 3.2$, $p = 0.0064$.

Methods

Power Analysis

Effect size was calculated using Daniel Lakens effect size spreadsheet, inputting the number of pairs and t-value into the dz from t for correlated samples section. G*Power was used to calculate the sample sizes needed for 95% power. t-tests were chosen as Test Family, "Means: difference between two dependent means (matched pairs)" as statistical test, a priori as the type of power analysis. The number of tails was two, the effect size was determined to be $dz = 0.826$, .05 was the error of probability. For 95% power, this yielded a sample of 22 participants, 18 participants for 90% power, and 14 for 80% power. A mistake in the first draft of this report indicated that only 21 participants was necessary for 95% power and this was set as the goal N prior to data collection.

Planned Sample

In order to have the greatest amount of power, and therefore be most likely to replicate the effect if it is present, we planned to include at least 21 participants in our sample. All participants included in this study were native English speakers. Half of the participants completed the object-naming task and half completed the color-naming task.

Materials

The original use of materials below was followed precisely:

"Four colors (red, green, blue, and pink) were used. For each color, we selected ten pictures (Alario & Ferrand, 1999) that shared at least the onset consonant with the color name. These color-object pairings were used for the congruent condition. The incongruent condition was created by repairing all pictures having a given onset with a particular color whose name had a different onset (e.g., all ''r'' pictures were re-paired with the same nonred color), so that picture names and color names did not overlap in phonology (see Table 1 for a list of the object names). Thus, there were 80 experimental items. An additional set of 80 filler items was created by pairing a set of 40 pictures with two different color names that did not overlap with the object names in phonology. Thus, overall, there were 40 congruent and 120 incongruent trials. Finally, eight of the filler items used in the experiment proper were also used during training. The original pictures were black line drawings on a white background; they were transformed into colored pictures by coloring the outline with one of the four designated colors.

Items were pseudorandomly ordered within four blocks of trials. Each block consisted of an equal number of experimental and filler trials, with each picture appearing once, and the different colors appearing an equal number of times. The same color never appeared on consecutive trials, and neither did pictures with a semantic or phonological relationship. There were never more than three items from the same condition in a row. The order of the blocks was counterbalanced following a balanced Latin-square design." (Janssen et al, 2008, p. 217-8).

Procedure

The original procedure below was followed precisely:

"Participants sat in front of a computer with an attached microphone and performed either the object- or the color-naming task. The experimental software was DMDX (Forster & Forster, 2003). For both tasks, participants first were familiarized with the objects and their names. On each trial, they saw a sequence consisting of a fixation cross (700 ms), a blank screen (200 ms), a picture (1,000 ms), and finally the picture with its name (1,000 ms). The appearance of the picture name cued participants to name the picture aloud. Participants in the color-naming task were then told the four color names that would be used in the subsequent two phases. After familiarization, participants were trained and then performed the experiment proper. These two phases had identical trial structures: a fixation cross (700 ms), followed by a blank screen (200 ms), followed by a picture of a colored object (1,500 ms). Depending on the instructions, participants named the object or the color (with the adjective un- marked for grammatical gender) upon picture presentation. There were eight trials in the training phase. The experiment lasted 25 min." (Janssen et al, 2008, p. 218).

Each response was digitally recorded to the computer hard drive to allow subsequent visual inspection of the response time trigger. Trials with inaccurate triggers were corrected using CheckVocal (Protopapas, 2007).

Analysis Plan

The original data treatment procedures and statistical analyses were followed precisely:

"Trial with errors, nonvocal responses, and no responses were discarded from the analysis. Reaction times (RTs) more than 3 standard deviations above or below a given participant's mean were also discarded." (Janssen et al, 2008, p. 218). Separate analyses with participants (F1) and items (F2) as random factors were conducted. The effects of interest were the difference in reaction times and errors between congruent and incongruent conditions for object naming and color naming, respectively. We planned to test this by performing within-subjects t-tests on reaction times and error rates from congruent and incongruent trials for object and color naming. We expected to find a significant difference in naming times in the object naming task, with shorter reaction times for congruent trials than incongruent trials. No differences were expected in the color naming task.

Differences from the Original Study

Participants in the present replication were recruited from the UK, while the original study was conducted in the United States. To accommodate vocabulary different between the two dialects of English, we replaced 3 experimental items (*glasses* > *goose*, *pitcher* > *peacock*, *pants* > *plate*). Additionally, in the original study RT triggers were not checked for accuracy as the CheckVocal software had not yet been developed. Other than these differences, there were no anticipated differences between the present replication and the original study.

Table 1: Experimental items used in this experiment. * indicates replacement items.

rabbit, rake, racket, rocket, ruler, rat, rope, record, ring, road, broom, bow, barn, barrel, bike, bottle, bear, bed, ball, book, glove, giraffe, guitar, grave, globe, ghost, goose*, goat, gun, glass, paddle, peacock*, parrot, peanut, plug, pear, pen, plate*, pig, pipe

(Post Data Collection) Methods Addendum

Actual Sample

To achieve an equal number of participants per task and per presentation list, we tested 24 participants in total, 12 per task. All participants were psychology students at the University of Dundee, Scotland. They received partial course credit for participating.

Results

Data Preparation

Each recorded response was examined for timing and naming accuracy, using CheckVocal (cf. Protopapas, 2007). Trials with errors, nonvocal responses, hesitations, and no responses were coded as errors and excluded from the analysis. Trials with premature triggers were reset manually. Error rates were very low. In the color naming task, 5 trials were excluded (out of 960). In the object naming task, error rates were higher, but still very low (47 out of 960, 4.8%). Because error rates were so

low, they were not analysed further. Trials that were 3 standard deviations from a subject's mean response time were coded as outliers and were also excluded from the analysis. In total, 12 outliers out of 960 trials were identified in the color naming task and 14 out of 960 trials were identified in the object naming task.

Confirmatory Analyses

Two sets of paired sample t-tests were conducted on the reaction time data with subjects (t_1) and items (t_2) as random factors. In the color naming task, color naming times in the congruent condition (RT = 635, SD = 105) were faster than in the incongruent condition (RT = 654, SD = 128), but this difference did not reach significance, t_1 (11) = 1.6, p = .149, d_z = .45; t_2 (39) = 1.76, p = .087, d_z = .28. The mean difference of 18.3ms was within the 95% confidence interval [-7.8, 44.4]. The observed power in the by-subjects analysis, obtained using G*Power, was .296.

In the object naming task, object naming times in the congruent condition (RT = 708.3, SD = 51.6) were faster than in the incongruent condition (RT = 720.6, SD = 38.7), but this difference only trended towards significance in the subjects analysis, t_1 (11) = 1.9, p = .09, d_z = .54, and did not approach significance in the items analysis, t_2 (39) = 0.9, p = .352, d_z = .15. The mean difference of 12.3ms was within the 95% confidence interval [-2, 26.6]. The observed power in the by-subjects analysis, obtained using G*Power, was .4.

Exploratory Analyses

Although no between-task comparisons were conducted in the original paper, we ran a 2 (task) by 2 (congruency) ANOVA to test for an interaction between congruency and task. Task was between subjects but within items. The interaction did not approach significance in either analysis, both F's < 1, p's > .5, confirming that the effect of congruency was comparable in both tasks.

Discussion

Summary of Replication Attempt

Following the results of Janssen et al (2008) Experiment 2, in this replication we expected to find no effect of congruency in the color naming task but a significant effect of congruency in the object naming task.

The results of this experiment failed to replicate Janssen et al (2008), Experiment 2. In both tasks, we found that naming was numerically faster in congruent compared to incongruent trials. Although our results trended in the expected direction, they did not reach statistical significance. It is worth noting, however, that in a preliminary analysis, which excluded error trials but did not yet exclude outliers, the results did partially replicate. Specifically, the observed trend in the subjects analysis just reached significance (p = .05), although the items analysis still failed to approach significance. Furthermore, Janssen et al predicted no effect of congruency in the color naming task, yet in our data the same numerical trend was observed.

Commentary

This replication used the original materials, experimental and filler, provided by the original authors (with the exception of 3 items, see above). It also used the

same procedure as described in the target article. The population was different; the original study used American university students, whereas the present replication used British university students. It is unlikely that cultural differences account for the failure to replicate as both populations have the same relevant syntactic constraints on word order.

While we did not find effects consistent with Janssen et al's predictions regarding language-specific word-order constraints, it is also important to note that we also did not observe any facilitation effect in the color naming task. Since publication of the target article, two published attempts to replicate this finding have been unsuccessful (Dumay & Damian, 2011; Madebach et al., 2011). However, both of these studies did observe phonological congruency effects in the color naming task, despite testing in languages with adjective+noun word order. More recently, further demonstrations of facilitation due to phonological congruency in color naming tasks have been reported (Macizo & Herrera, 2014, but see Roux, Bonin, & Kandel, 2014 for mixed results). The fact that we found no congruency effect in either task is potentially problematic for this replication. It should further be noted that the power analysis conducted on the basis of the target article suggested that only 21 participants (10.5 per task) would be needed to achieve 95% power in this replication. Compared to other studies on this topic, this N is very small; sample sizes in comparable experiments range from 18-50 participants per task. Therefore, the small sample tested here and the failure to observe any congruency effect may suggest that this replication was underpowered.

Finally, this replication used CheckVocal (Protopapas, 2007) to manually inspect responses for early or late voice key triggers; these trials were corrected. The original study, which was run prior to the development of this software, was not able to inspect voice key accuracy on individual trials. This could have influenced the outcome of the study in one of two ways. First, more trials may have been excluded as errors. The original report excluded 3.8% and 3.5% of trials in the color and object naming tasks, respectively (compared with 0.5% and 4.8% errors in the color and object naming tasks, respectively in this replication). Secondly, some voice key errors may have been included in the final analysis. Given the influence that removing 14 extreme values had on the results in this replication, it is clearly possible that the effect could depend on the inclusion of error trials.

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